Geometrical interpretation of vector product

Construct a parallelogram $OACB$ with $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$ as adjacent sides.

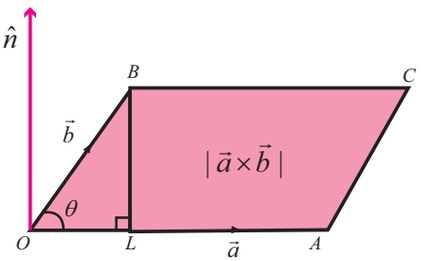


Fig. 8.44

Let $|\underline{AOB} = \theta$

From the diagram,

$$\sin\theta = \frac{BL}{OB}$$

$$BL = (OB)\sin\theta = |\vec{b}|\sin\theta$$

Now $|\vec{a}\times\vec{b}| = |\vec{a}||\vec{b}|\sin\theta = |\vec{a}|(BL)$

= (base) (height) = area of the parallelogram $OACB$.

Thus, $\vec{a}\times\vec{b}$ is a vector whose magnitude is the area of the parallelogram, having $\vec{a}$ and $\vec{b}$ as its adjacent sides and whose direction $\hat{n}$ is perpendicular to the plane containing $\vec{a}$ and $\vec{b}$ such that $\vec{a},\vec{b},\hat{n}$ form a right handed system.

Thus, $|\vec{a}\times\vec{b}|$ = area of the parallelogram whose adjacent sides are $\vec{a}$ and $\vec{b}$.

From the area of the parallelogram, we can deduct the area of the triangle OAC as half of the area of $OACB$.

Deduction

The area of any triangle whose two sides are $\vec{a}$ and $\vec{b} = \frac{1}{2}|\vec{a}\times\vec{b}|$.

8.8.5 Properties

(i) Vector product is non-commutative

By definition

$$\vec{b}\times\vec{a} = |\vec{b}||\vec{a}|\sin\theta(-\hat{n})$$

[since $\vec{b},\vec{a},-\vec{n}$ form a right handed system]

$$= -|\vec{a}||\vec{b}|\sin\theta\,\hat{n}$$

$$= -(\vec{a}\times\vec{b})$$

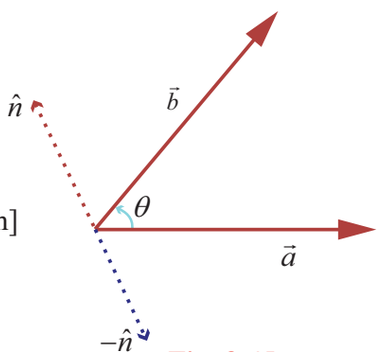


Fig. 8.45

Thus vector product is non-commutative.

(ii) If two vectors are collinear or parallel then $\vec{a}\times\vec{b} = \vec{0}$ (how?)

But $\vec{a}\times\vec{b} = \vec{0} \Rightarrow$ $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\vec{a}$ and $\vec{b}$ are parallel.

(iii) For any two non-zero vectors $\vec{a}$ and $\vec{b}$, $\vec{a}\times\vec{b} = \vec{0} \Leftrightarrow \vec{a}$ and $\vec{b}$ are parallel.

Deduction

$$\vec{a}\times\vec{a} = \vec{0}$$

(iv) With usual meaning of $\hat{i},\hat{j}$ and $\hat{k}$ (they form a right handed system), the following results are obtained.

It is clear that,

$$\hat{i}\times\hat{i} = \hat{j}\times\hat{j} = \hat{k}\times\hat{k} = \vec{0}$$

$$\hat{i}\times\hat{j} = \hat{k}\ ;\ \hat{j}\times\hat{k} = \hat{i}\ ;\ \hat{k}\times\hat{i} = \hat{j}$$

$$\hat{j}\times\hat{i} = -\hat{k}\ ;\ \hat{k}\times\hat{j} = -\hat{i}\ ;\ \hat{i}\times\hat{k} = -\hat{j}$$ (how?)

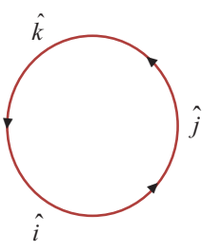


Fig. 8.46

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